

Joint Compute-Control Budgeting for Receding-Horizon Diffusion Policy

Iso-Compute Allocation of Denoising and Replanning for Efficient Robotic Manipulation

Abstract

Diffusion Policy uses iterative action denoising and receding-horizon execution. This project reproduces the official low-dimensional Push-T checkpoint with the upstream workspace and runner, obtaining a 50-seed mean score of 0.919 in the current compatibility environment. It then evaluates how to allocate a fixed inference budget between denoising quality and replanning frequency in a deterministic Push-T-style surrogate.

Problem

We define k as denoising steps per policy call, h as executed controls before replanning, and $B \sim k/h$ as amortized compute per control step. The grid evaluates k in $\{2,4,8,16\}$ and h in $\{1,2,4,8\}$, with main analysis on $B=1$, $B=2$, and $B=4$ frontiers.

Official Reproduction

The official source was acquired through a GitHub code load archive after git clone instability, and the 1044185793-byte Push-T checkpoint was downloaded from the project server. The 50-seed official runner result is 0.919; the checkpoint filename reports 0.969, so the difference is recorded as an environment-version reproduction gap.

LeRobot Supplement

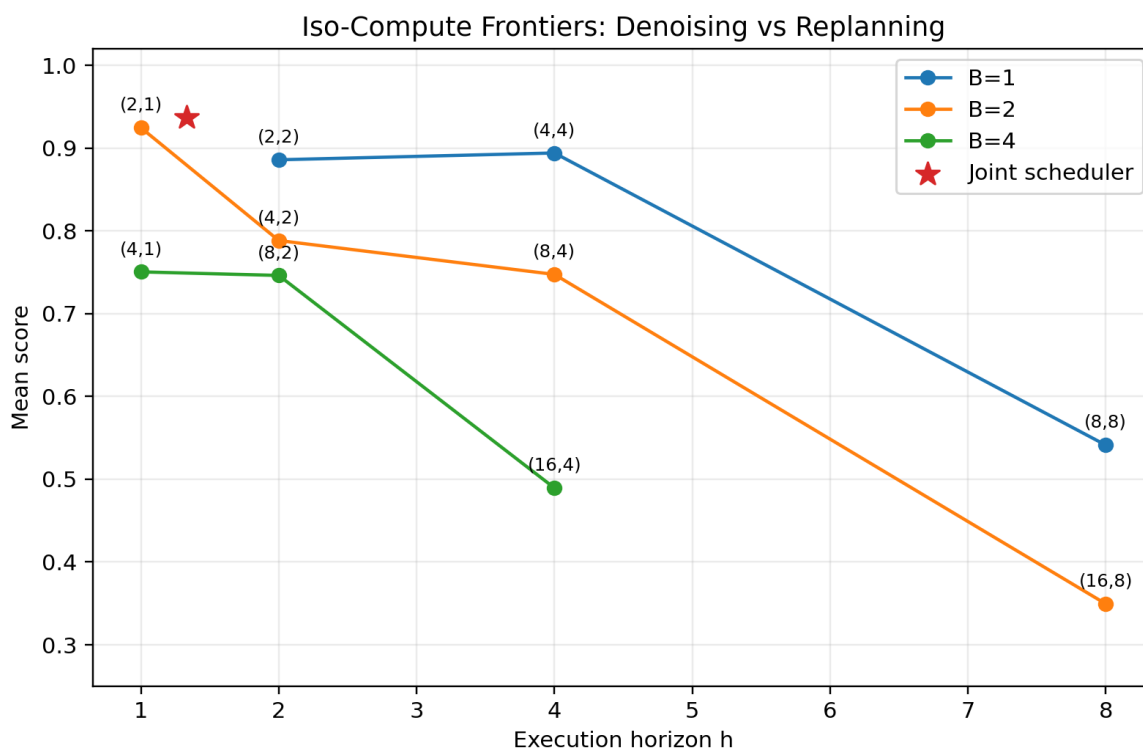
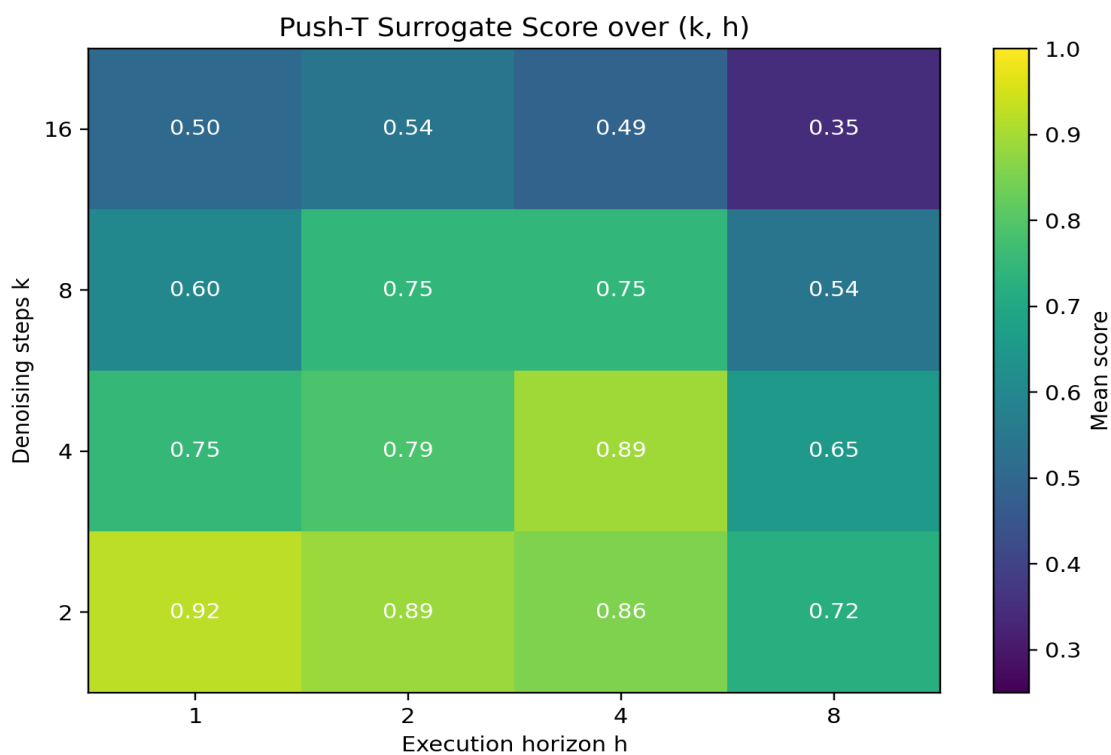
The LeRobot lerobot/diffusion_pusht model card reports 500-episode average max reward 0.955 and success 65.4%. The local mirror-downloaded checkpoint loads and runs in gym-pusht; a 300-step, 100-denoising-step smoke test reached max reward 0.627. Because the file SHA did not match HF LFS metadata, local rollout results are marked as smoke verification.

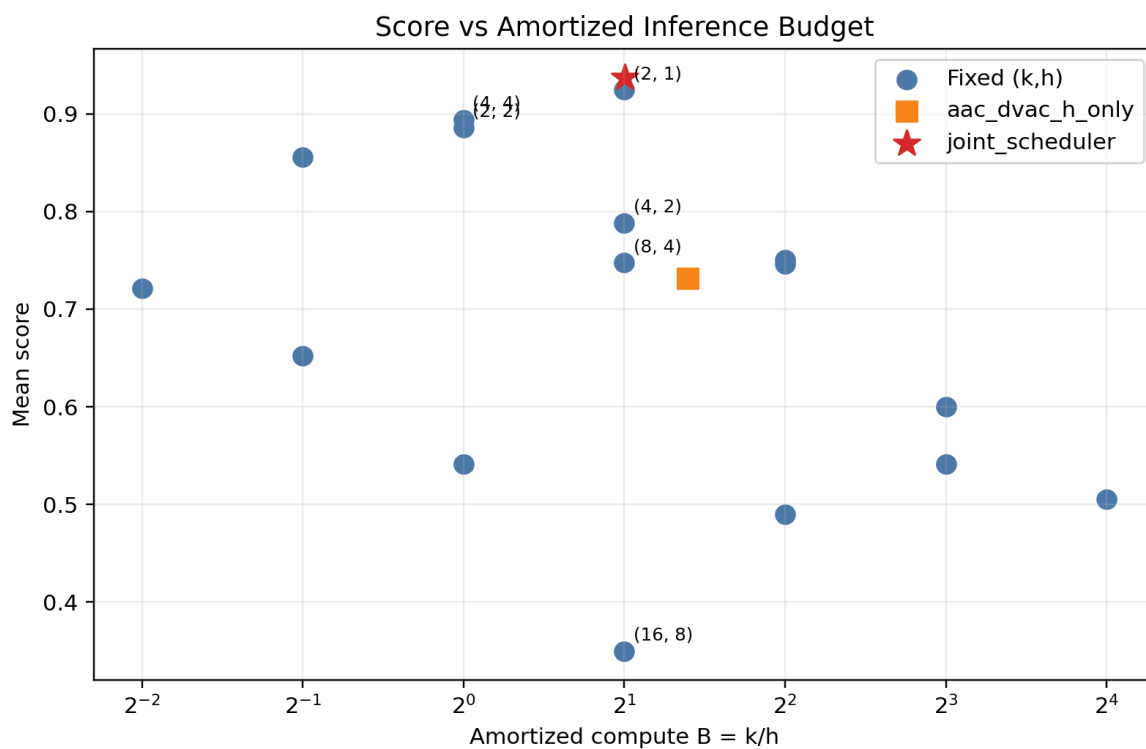
Method

A fixed runner evaluates every (k,h) . A training-free joint scheduler operates on the $B=2$ frontier. It selects (4,2) in free-space approach and final alignment, and (2,1) during close-contact pushing or unstable progress.

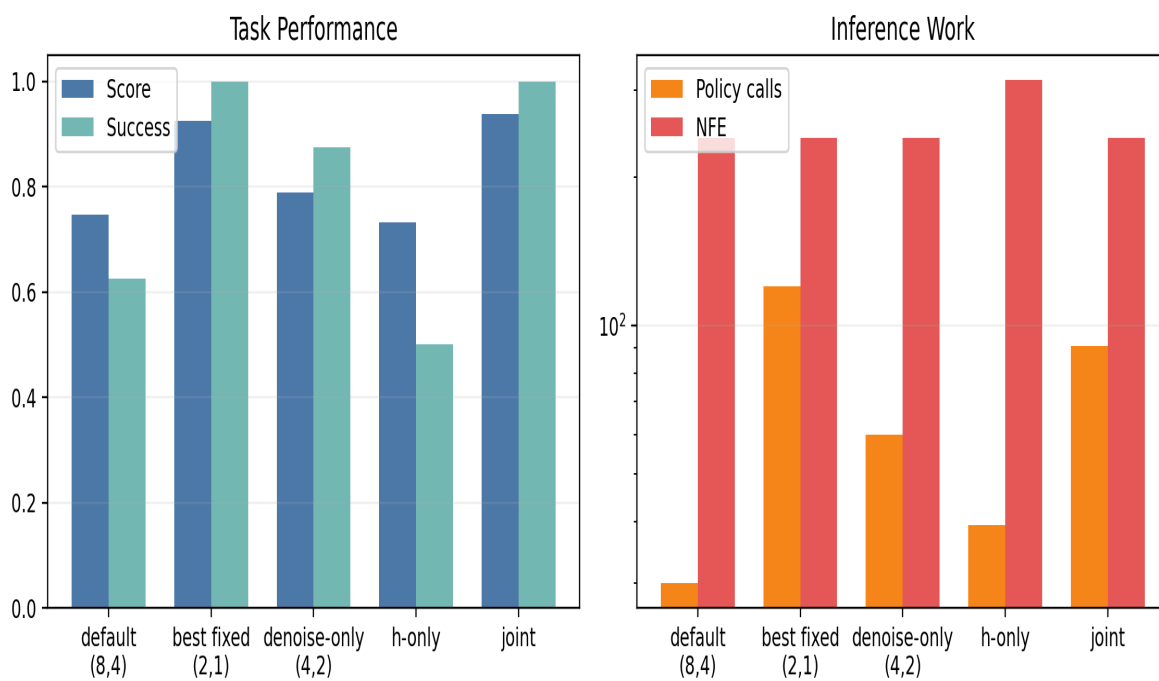
Main Comparison

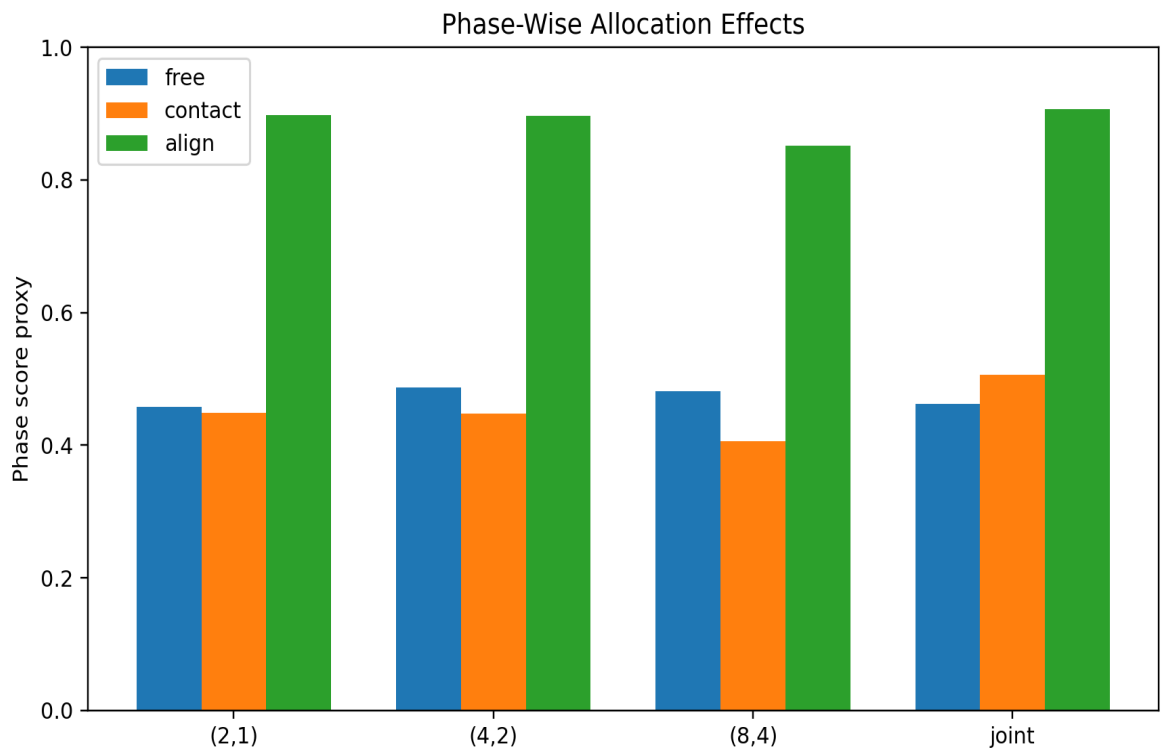
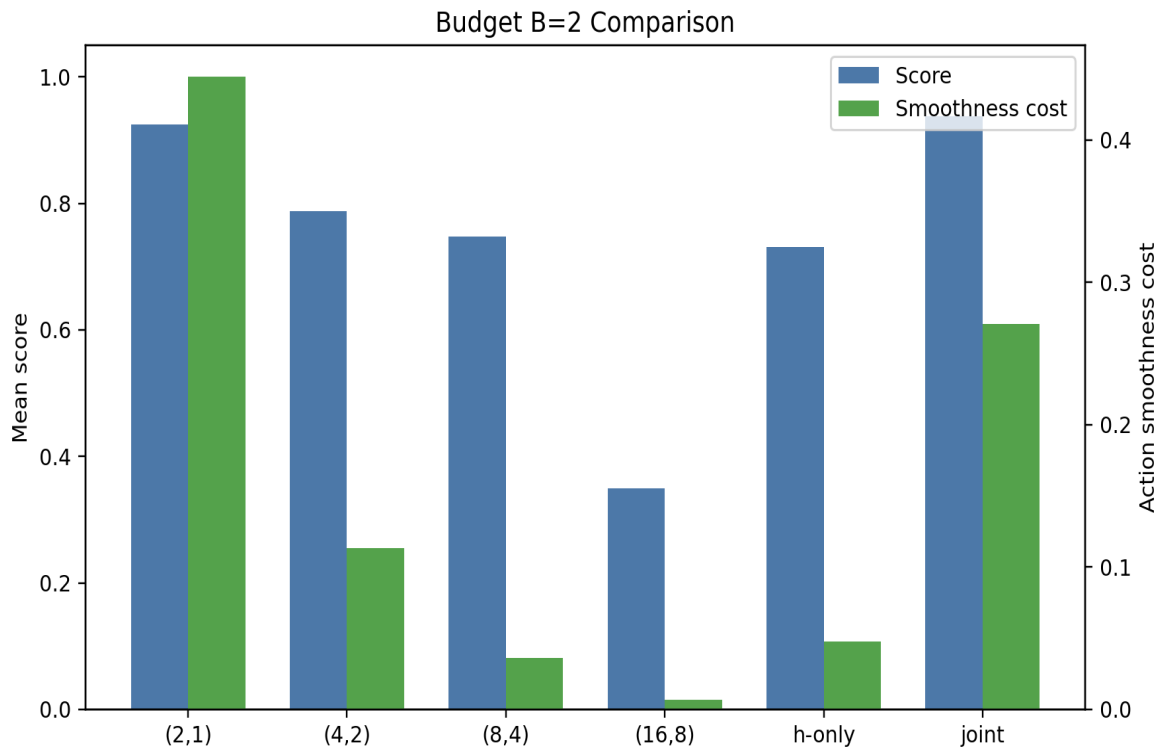
Method	B	Score	Success	Calls	NFE	Smoothness
Default DP-style (8,4)	2.00	0.747	0.625	30.0	240.0	0.036
Best fixed B=2 (2,1)	2.00	0.925	1.000	120.0	240.0	0.444
Denoising-only point (4,2)	2.00	0.788	0.875	60.0	240.0	0.113
Long denoise/chunk (16,8)	2.00	0.350	0.125	15.0	240.0	0.007
AAC/DVAC-style h-only	2.62	0.731	0.500	39.4	315.0	0.048
Joint scheduler	2.01	0.937	1.000	90.8	240.8	0.271





Main Baseline Comparison under $B \approx 2$





Results

The joint scheduler reaches score 0.937 at average $B=2.006$, compared with 0.925 for the best fixed score baseline (2,1). It reduces policy calls from 120.0 to 90.8 and smoothness cost from 0.444 to 0.271.

Limitations

The scheduler-improvement numeric results are from the surrogate, not official Diffusion Policy checkpoint rollouts. The official checkpoint run uses compatibility patches for Python 3.12 / Gym 0.26, and the LeRobot mirror checkpoint is executable locally but not SHA-verified. The compute model is approximate, the scheduler is rule-based, and the experiment is limited to one task family.

Conclusion

The experiment supports the hypothesis that denoising and replanning should be allocated jointly. Contact phases favor frequent replanning; easier phases can use a longer chunk and more denoising while staying on the same compute frontier.

References

Diffusion Policy project page: <https://diffusion-policy.cs.columbia.edu/>. Official code: https://github.com/real-stanford/diffusion_policy. LeRobot model card: https://huggingface.co/lerobot/diffusion_pusht. Paper: Diffusion Policy: Visuomotor Policy Learning via Action Diffusion, RSS 2023 / arXiv 2303.04137.